

THE WORLD ENERGY CRISIS IN PERSPECTIVE:
THE BREAKTHROUGH TECHNOLOGY THAT SOLVES IT QUICKLY

© T. E. Bearden
Lt. Col., U.S. Army (Retired)
M.S. Nuclear Engineering

INTRODUCTION:

The seriousness of the world energy crisis demands the development of a breakthrough new technology to solve it -- and simultaneously to dramatically clean up the biospheric pollution presently ongoing by most of our present rather "dirty" energy sources and applications. The necessary technology is already spelled out and waiting; it just needs doing with a determined scientific effort, well-staffed and well-funded. But it is a PHYSICS problem, not an "electrical engineering" problem. Indeed, the first prerequisite is to recognize the deliberate curtailment of the EE model that has brought on and produced this crisis. We spell it out in this short synopsis paper.

We wish great success to anyone tackling this urgent problem, and we do salute all serious efforts toward solving it, so long as they are properly oriented on PHYSICS and not just still bound and shackled by the highly crippled electrical engineering. Tackling the problem from the standard EE approach guarantees failure from the outset! Tackling it from the known modern physics level guarantees success from the outset!

Our purpose is to inform the reader of some of the major barriers one faces in achieving the breakthrough technology, because of the present horribly mutilated and crippled state of the old electrical engineering (EE) model (from the 1880s. The EE (Heaviside-Lorentz) model was deliberately maimed and mutilated in 1892 by Lorentz's further modification of Heaviside's vector equations to deliberately symmetrize them. The continued use and universal acceptance of this archaic and severely restricted EE model is the unsuspected reason for the world energy crisis and much of the atmospheric pollution by fueling our present energy sources.

Because of this unrecognized crippled state of our present electrical engineering (and thus of our present electric power engineering), the usual "new energy" announcement or effort is a sad joke, as is most of the present world's scientific policy on "solving" energy-related matters. As a certified example, our own Department of Energy already has operating COP = 200% to 700% experimental solar cells (see Klimov et al.) that produce a freely amplified electron output current, using excess EM energy input extracted directly from the local seething virtual state vacuum. This work has been widely published in leading physics and nanocrystalline journals, and it has been experimentally demonstrated and proven by two great U. S. National Laboratories. Thus it has been solidly proven scientifically, forever. More on that later, including rigorous scientific references.

Several outstanding inventors (particularly such as John Bedini) also have patented energy and power systems (demonstration models) that prove the NEW PHYSICS approach to electrical power, and totally validate its ability to solve the crisis. However, for almost a century these inventors have been harassed, denied funding, denigrated, and some have even been killed. The standard objection -- that free EM energy systems are impossible -- is based on the wide assumption that the sad old 1892 EE model is

"perfect", and that it already covers everything that can be known about electrical power. This is particularly ironic since adherence to the present grossly inadequate EE model and practice is directly responsible for the world energy crisis in the first place!

In this informal summary, we will review what happened to develop electrical engineering itself, and where and how it went wrong.

The present electrical engineering (and hence electric power) model was kluged together decades before the advent of modern physics such as special and general relativity, quantum mechanics, quantum field theory, gauge field theory, etc. and indeed mostly before the discovery of the electron. In this paper we present some little-recognized facts regarding electric power and EM energy. These facts are outside the present archaic electrical engineering model and symmetrized technology, so they require that a physicist (and preferably one who knows group theory and modern quantum physics) look at them, not an electrical engineer whose curriculum does not include group theory and whose model specifically excludes the exact Maxwellian energy systems that are required to solve the problem.

1. NIKOLA TESLA HAD ALREADY DISCOVERED FREE ENERGY CIRCUITS BEFORE ELECTRICAL ENGINEERING WAS EVEN BORN.

FIRST A LITTLE HISTORY:

Maxwell died in 1879, and very promptly persons such as Gibbs and Heaviside jumped in and tore apart the hated quaternions, producing the far more limited vectors and vector algebra itself. Tesla of course gave the world successful AC power, which could be transmitted over mainlines for large distances, making it practical. But electrical engineering was not yet born; instead, there were some three dozen or so PHYSICISTS worldwide who also knew something of Maxwell's quaternion-like EM theory and model. As AC power began, it became obvious that a new kind of engineer was needed -- one who was trained on the new electromagnetic systems and who could perform the necessary work in designing, emplacing, and maintaining these new power systems. Thus arose the need for electrical engineering, leading to its deliberate formation and being introduced into our universities.

In the early 1890s, the model to be used by the new "electrical engineering" was being put together/determined by H. A. Lorentz, and initially he had tentatively selected the simplified vector Heaviside model as the model to be utilized. We specifically note that the original Heaviside vector theory is still ASYMMETRIC, even though simplified -- and thus it still contains EM systems which can freely output more energy in their forward EMF/MMF region to power their loads usefully, than they output into their back EMF/MMF region to unwittingly destroy the free flow of energy (from the vacuum) in the system itself. In short, Heaviside's original model still includes Maxwellian systems that can output more energy in usefully powering the load than the operator has to input and pay for. As we shall see, the operator does not furnish the energy to power the loads and losses, but only to continually restore the source dipole inside the generator.

We further point out that a SYMMETRIC system, to which present EE is restricted, does not contain these asymmetric Maxwellian COP>1.0 systems that Heaviside's original equations/model still contained.

In 1892, the Heaviside vector electrical engineering (EE) mathematics model was deliberately mutilated and horribly crippled by Lorentz -- by his deliberately altering the equations to symmetrize them -- just before the birth of EE itself in our universities. **We strongly stress:** It is indeed possible to build asymmetric Maxwellian/Heaviside EM circuits and systems that by definition can receive and use extra EM energy freely from the active medium (the active vacuum). As a simplest possible example, every charge and dipole is already such a "free energy system" freely absorbing energy from the virtual state vacuum and radiating it continuously as real, observable EM energy. Shortly we will cite the rigorous proof.

But first to put the history into perspective:

The early physicist Nikola Tesla gave us AC power, the rotating magnetic field that made modern AC motors and generators possible, and radio. Contrary to what is taught in most of our texts, Marconi did not invent radio; instead, he stole it from Tesla's previous work. Sir John Ambrose Fleming, noted English scientist who studied electricity and magnetism under Maxwell and who was also an electrical advisor of the Edison Electric Light Company of London from 1881 to 1891 or so, was also an advisor to the Marconi Company from 1899. In 1901, e.g., Fleming was also advising Marconi in trying to bypass Tesla's actual patents and prior art. [See Sungook Hong, Wireless: From Marconi's Black-Box to the Audion, MIT Press, 2001, p. 72. "In his letter to Marconi on February 19, 1901, Fleming reassured him by saying that Tesla could not do anything, and that 'if you can receive there [in America], you will establish priority'". Marconi followed this advice and obtained patents. When Marconi later tried to bring lawsuit against the US Army Signal Corp and claim that their use of radio violated Marconi's patent rights, Marconi lost the suit. Tesla had already demonstrated priority of art, and this was specifically upheld by the U.S. Supreme Court in 1943, shortly after Tesla's death.

Further, in the late 1880s and early 1890s, the amazing Tesla had already discovered self-powering "free energy from the vacuum" electrical systems and circuits (Tesla's term for the vacuum was "the active medium"). Accordingly, Tesla was already briefing leading technical groups -- such as the American Institute of Electrical Engineers -- that humanity could freely obtain all the EM energy it wished and needed, freely and directly from the active medium itself, without the consumption of any fuel at all. Here are some direct quotes from Tesla:

"Ere many generations pass, our machinery will be driven by a power obtainable at any point in the universe. This idea is not novel... We find it in the delightful myth of Antheus, who derives power from the earth; we find it among the subtle speculations of one of your splendid mathematicians... Throughout space there is energy. Is this energy static or kinetic? If static our hopes are in vain; if kinetic -- and this we know it is, for certain -- then it is a mere question of time when men will succeed in attaching their machinery to the very wheelwork of nature." [Nikola Tesla, in a speech in New York to the American Institute of Electrical Engineers, 1891. Quoted from back cover of his biography, Margaret Cheney, Tesla: Man Out of Time, Simon and Schuster, 2001]. [Note particularly that the very next year, 1892, Lorentz was elicited by J. P. Morgan's scientists to deliberately alter the still-asymmetric Heaviside equations so that the new Heaviside-Lorentz model was symmetric -- which meant that all such "Tesla overunity systems" were deliberately omitted from the new electrical engineering just being born.]

"Electric power is everywhere present in unlimited quantities and can drive the world's machinery without the need of coal, oil, gas, or any other of the common fuels." [Nikola Tesla].

"We have to evolve means for obtaining energy from stores which are forever inexhaustible, to perfect methods which do not imply consumption and waste of any material whatever. I now feel sure that the realization of that idea is not far off. ...the possibilities of the development I refer to, namely, that of the operation of engines on any point of the earth by the energy of the medium..." [Nikola Tesla, during an address in 1897 commemorating his installation of generators at Niagara Falls.].

"Whatever our resources of primary energy may be in the future, we must, to be rational, obtain it without consumption of any material." [Nikola Tesla, 1900].

For the rigorous mathematical proof that some of Tesla's actual patented circuits could already do this, see T. W. Barrett, "Tesla's Nonlinear Oscillator-Shuttle-Circuit